Section 6 — N(Z): The Normality-Defect Index ("ZEUS")

Status: HYPOTHESIZED tier. Not asserted as isomorphic to any Pellis construct. Proposed as a single bounded dial that gives the existing eigenvector-condition-number / Henrici departure-from-normality discussion (§4) a normalized index, with the rotation (normal-operator) regime and the exceptional-point (defective-operator) regime as its two literal poles.

6.1 Motivation

§4 established that the leverage operator J moves between two qualitatively distinct regimes:

- **Normal regime:** J normal ($JJ^* = J^*J$) — e.g. the pure-rotation case $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$. Eigenvalues $e^{\pm i\theta}$; eigenvectors $(1, \mp i)/\sqrt{2}$ — fixed, orthogonal, independent of θ. Eigenvector condition number $\kappa(V) = 1$.
- **Defective limit:** J at the exceptional point ($\gamma_1\gamma_2 = -(\epsilon - \delta)^2$) — eigenvalues coalesce ($\lambda_+ = \lambda_-$), eigenvectors collapse parallel, geometric multiplicity drops to 1, $\kappa(V) \to \infty$.

These are not two arbitrary reference points — they are the two structural poles of operator theory itself (normal vs. maximally non-normal). Naming them as fixed anchors of a bounded index is a definitional move, not a claim that they converge.

6.2 Definition

Let $\Delta(J)$ be the Henrici departure-from-normality:

$$\Delta(J) = \sqrt{\|J\|_F^2 - \sum_i |\lambda_i|^2} \;\geq\; 0$$

$\Delta(J) = 0$ exactly when J is normal. Define the bounded, signed **N(Z) index**:

$$N(Z) \;=\; 1 \;-\; \frac{2\,\Delta(J)}{\Delta(J) + \|J\|_F} \;\in\; (-1,\, 1]$$

Pole	Operator state	Eigenvector geometry	$\kappa(V)$	$N(Z)$
$N = +1$	Normal (rotation regime)	Orthogonal, angle-invariant	1	$+1$
$N \to -1$	Maximally defective (exceptional point)	Parallel / coalesced	$\to \infty$	$\to -1$

6.3 "ZEUS" — naming convention for the index

Zipper Energy United Spectrum (ZEUS) names $N(Z)$ as an object, with each term keyed to a specific feature of the operator, not used as independent evidence of a mechanism:

Term	Referent	Basis
United **Spectrum**	$\lambda_+ = \lambda_-$ at $N \to -1$ — eigenvalues literally coalesce	Exact (derived in §3)
Energy	$\mathrm{Re}(\lambda)$, dissipation/decay term carried from the REAL mapping (§5)	Hypothesis-tier, inherited
Zipper	Eigenvector state: orthogonal/distinct at $N = +1$ ("open"), parallel/merged at $N \to -1$ ("closed")	Interpretive label for a derived mechanism — naming convention, not independent proof

Note the asymmetry: "United Spectrum" names the $N = -1$ pole precisely; it does not separately name the $N = +1$ pole (maximally *split*, orthogonal spectrum). The name foregrounds the collapse end of the dial. This is consistent with naming conventions elsewhere in canon (SIGNET, SCALULAR) and does not affect the validity of $N(Z)$ itself — it is a property of the label, not the index.

6.4 Relationship to existing constructs

$N(Z)$ is **not** proposed as a replacement for $\kappa(V)$ or $\Delta(J)$ — both remain the primary, unbounded diagnostic quantities (§4). $N(Z)$ is a normalized presentation layer for communicating where a given instrument's operator sits on the normal ↔ defective spectrum, bounded for comparison across instruments with different scales of $\|J\|_F$.

Open items (pending Pellis):

1. Whether $\Delta(J)$ (vs. $\kappa(V)$ directly) is the correct quantity to normalize for this purpose.

2. Whether $N(Z) = -1$ should be reserved strictly for the exceptional point, or extended to the broader defective neighborhood around it.

Drafted 2026-06-21 for insertion into the ERES–Pellis spectral bridge document, following §5 (Proposed Mapping). HYPOTHESIZED tier throughout, consistent with document-wide status convention.